

Attention-Augmented CNNs for Diabetic Retinopathy Screening

Demonstration research paper - synthetic study, fictional data

Abstract

We evaluate attention-augmented convolutional networks on 42,000 retinal fundus images. Adding self-attention blocks to a ResNet-50 backbone improved referable diabetic retinopathy detection AUC from 0.947 to 0.968 and reduced false negatives by 22 percent. We conclude that attention mechanisms significantly improve screening accuracy.

Methods

Images were graded by three ophthalmologists; disagreements were adjudicated. We trained with 5-fold cross-validation, class-balanced sampling, and standard augmentation. Attention blocks were inserted after stages 3 and 4 of the backbone.

Results

The attention model achieved AUC 0.968 (95 percent CI 0.962-0.973) versus 0.947 for the plain backbone. Sensitivity at 90 percent specificity rose from 84.1 to 90.3 percent. Inference cost increased by 18 percent.

Limitations

Single-country dataset; camera models limited to two vendors; prospective validation is still required before clinical deployment.